Programming with Blockly

In the simulator first. Simple things first.

Every program starts under **when RUN is pressed**. Press **RUN** to try it, **STOP** to stop it, **Back to start** to put the robot back. Mistakes are fine: change one thing and run it again.

1 Open Blockly

On a computer: robotics.makerlabkids.com/mecanum/blocks. Pick **Simulator**. The pretend robot is on the right.

2 Your first move



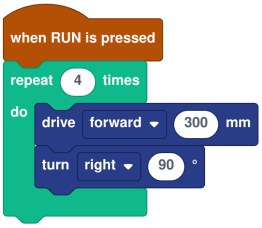
Drag **drive forward** from **Moves** and snap it under **when RUN is pressed**. Press **RUN**.

3 Add a turn



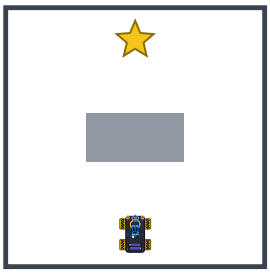
Snap **turn right 90°** under it. Click a number to change it. Run it again.

4 Repeat



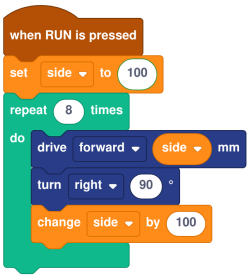
A square is 4 times "drive, turn". **repeat** (in **Loops**) does it for you.

5 Pick a map



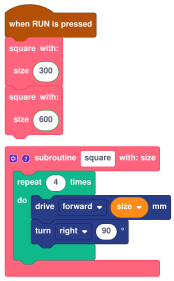
Under the simulator: **Map**. Try **2. Star behind the box**. Drag the robot to move where it starts.

6 Variables



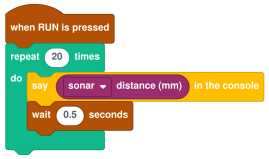
A variable is a box with a number in it. **change side by 100** makes each side longer: a spiral.

7 Subroutines



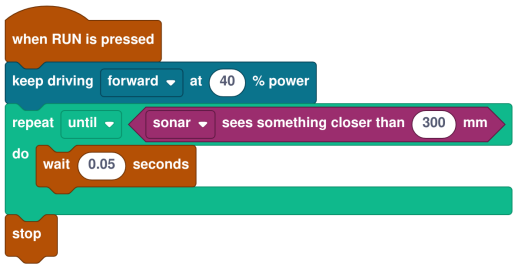
Make your own block (**Subroutines**). The gear adds an input like **size**.

8 Watch a sensor



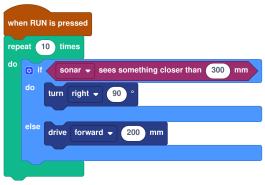
say puts a number in the **Console**. Watch the sonar change as the robot moves.

9 Repeat until



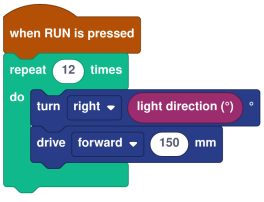
Drive until something is near, then stop. The wall is somewhere new every run on **map 4**.

10 If ... else



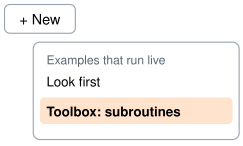
Decide! If the way is blocked, turn. Else, go forward. Try **maps 7 and 8**.

11 Follow the light



light direction says which way the lamp is. Turn that much. Try **maps 6 and 9**.

12 The toolbox



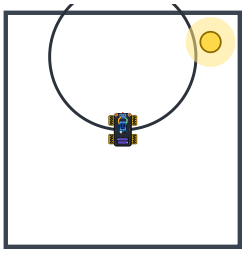
+ New, then **Toolbox: subroutines**: ready-made blocks like **creep to wall**. On the real robot: **Bluetooth**, **Robot**, **RUN**.

Ten maps

Pick one under the simulator: **Map**. The star says when you made it.

Maps **4 to 10 change every run**: the wall moves, the door opens and closes, the lamp goes dark. A program that only counts millimeters will miss. Use the **sensors** and **if** to decide.

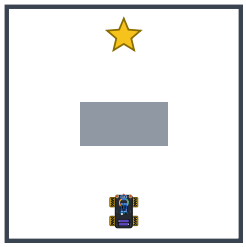
1 Square room



Drive a square. Then try the light and the loop of tape.

Use: drive, turn, repeat

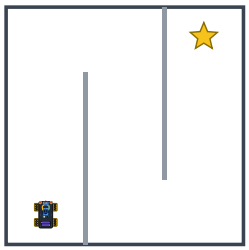
2 Star behind the box



Get to the star. Go around the box.

Use: drive (slide left / right), turn

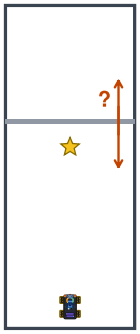
3 The hallway



Follow the hallway to the star. Plan every move.

Use: drive, turn, a subroutine for each part

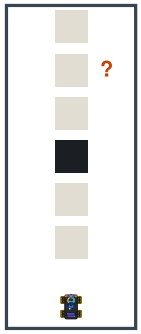
4 Stop at the wall



The wall moves every run. Stop on the star, closer than 20 cm, no bump.

Use: keep driving, repeat until, sonar sees closer than

5 Stop on the dark tile



A different tile is dark every run. Stop on it.

Use: repeat until sees the line, stop

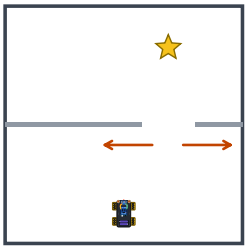
6 Find the light



The lamp is in a different corner every run. Find it.

Use: light direction, turn, follow the light

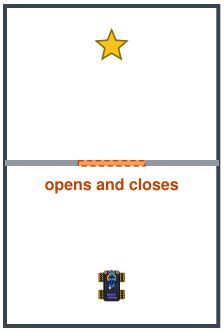
7 Find the gap



The gap moves every run. Slide until the sonar sees through it, then go.

Use: keep driving (slide), repeat while, sonar

8 The door



Wait until the sonar sees the door open, then go through.

Use: repeat while, wait, sonar, drive

9 Red light, green light



Move only while the lamp is on. Moving in the dark gets you caught.

Use: light level, if ... else, wait

10 The maze



Look before you go: turn, check the sonar, pick the open way.

Use: subroutines, if, sonar, repeat

Made it? The Console says **Made it!** with your time, bumps and catches. Run it again: does it still work?

Stuck? Put **say** blocks in your program to see what the sensor reads, and slow the simulator to **real time**.

Need a hand? + New, **Toolbox: subroutines**: copy a ready-made subroutine into your tab (Ctrl+C, Ctrl+V).